ETHAN KNOX

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SUMMARY

Applied mathematician completing an M.S. in Applied Mathematics at UIUC (May 2026), with a strong foundation in statistics, probability, and large-scale numerical computation. Experienced building Python pipelines that extract quantitative insight from large, complex datasets, and translating mathematical models into validated, computable implementations. Fluent across the scientific Python stack and GPU-accelerated data workflows. Seeking a data or quantitative analyst role.

EDUCATION

University of Illinois — Urbana-Champaign Champaign, IL
M.S. in Applied Mathematics *Aug 2024 – May 2026*

Purdue University West Lafayette, IN
B.S. in Mathematics (Minor: Psychology) • B.S. in Physics (Minor: Astronomy) *Jun 2017 – May 2021*

Honors: Dean's List; Semester Honors; Provost Summer Scholarship; Frank O'Bannon Honors & Accelerated Scholarships.

EXPERIENCE

Independent Researcher — Quantitative & Computational Modeling May 2023 – Present
Self-directed *Champaign, IL*

- Built and numerically **validated quantitative simulation models**, including a GPU-accelerated Monte Carlo engine benchmarked against closed-form analytical results — emphasizing data validation and reproducibility.
- Developed **Nulltracer** (Python / CuPy / CUDA), a GPU-accelerated numerical engine processing large-scale array computation with five high-order integrators — an exercise in performance optimization and large-data processing.

Graduate Research Assistant — Kirr Research Group, Dept. of Mathematics Aug 2025 – Dec 2025
University of Illinois — Urbana-Champaign *Champaign, IL*

- Co-authored a research report applying probabilistic and asymptotic analysis to a complex, high-dimensional mathematical model.
- Reduced an intractable model to a tractable, computable system using dimension-reduction techniques; derived and verified analytical results numerically.

Undergraduate Research Assistant — Iyer-Biswas Lab Aug 2020 – Dec 2020
Purdue University *West Lafayette, IN*

- Built custom **Python pipelines to extract, clean, and quantify measurements** from large experimental imaging datasets, contributing to published research on stochastic single-cell behavior.
- Applied statistical and stochastic-process analysis to experimental data; presented findings and edge-case analysis in weekly all-staff meetings.

SKILLS

Data & Analysis: Statistical modeling, probability & statistics, data processing & cleaning, Monte Carlo simulation, data visualization, numerical methods, large-dataset analysis

Programming & Tools: Python (Pandas, NumPy, SciPy, Matplotlib, SymPy, CuPy), C/C++, CUDA / GPU computing, Git, LaTeX, Linux / Bash

ADDITIONAL

Eagle Scout, Boy Scouts of America • Brazilian Jiu-Jitsu practitioner